

Josiane Uwineza

Data Scientist & Machine Learning Engineer — Medical Image Analysis

Kigali, Rwanda | +250 781 973 094 | inezajoyorange@gmail.com

Google Scholar | GitHub | LinkedIn

Research Interests

Deep learning for medical image analysis, specializing in ultrasound and CT workflows (2D/3D classification, object detection, segmentation, and multimodal registration). My primary doctoral research objective focuses on algorithmic robustness, causality, and model generalizability across disparate scanner vendors, clinical sites, and patient populations in low-resource settings.

Education

MSc, Control Science and Engineering

2017 – 2019

Beijing Institute of Technology, China

- Thesis: “Static Hand Gesture Recognition for Human–Robot Interaction.”

BEng, Electrical Engineering

2012 – 2016

University of Rwanda, College of Science and Technology

Publications

1. **Uwineza, J.**, et al. “Development and Temporal Validation of a Deep Learning Model for Automatic Fetal Biometry from Ultrasound Videos.” *Journal of Gynecology Obstetrics and Human Reproduction/Elsevier Masson*, 2025.
2. **Uwineza, J.**, et al. “Towards Automatic Verification of the Critical View of the Myopectineal Orifice with AI.” *Surgical Endoscopy/Springer US*, 2023. (EI-indexed).
3. **Uwineza, J.**, et al. “Static Hand Gesture Recognition for Human–Robot Interaction.” *International Conference on Intelligent Robotics and Applications/Springer International*, 2020. (EI-indexed).
4. **Uwineza, J.**, et al. “Malaria Disease Prediction Based on Machine Learning.” *2019 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)/IEEE*, 2020. (EI-indexed).

Professional & Research Experience

Research Institute Against Digestive Cancer (IRCAD Africa)

Kigali, Rwanda

Senior Machine Learning Engineer & Data Scientist

Jan 2023 – Present

- **Annotation Review Tooling for 3D Slicer** (Aug 2026 – Present): Leading development of a 3D Slicer extension that lets a second expert review segmentation quality directly on the images, replacing a manual spreadsheet workflow with located issues, comment threads, and reviewer suggestions.
- **Clinical Validation of Fetal Biometry Models** (Aug 2025 – Jul 2026): Validated fetal standard-plane classification and biometry models on a prospectively acquired clinical trial dataset in Kigali, Rwanda. Diagnosed and resolved persistent abdominal circumference plane recognition failures through a systematic annotation quality review, raising AC plane classification accuracy from 0.41 to 0.80.
- **3D Multimodal Image Registration** (Mar – Jul 2025): Developed a rigid/non-rigid vessel registration pipeline aligning 3D ultrasound and CT volumes via extracted centerlines. Implemented

and evaluated Go-ICP and RANSAC algorithms, utilizing Open3D and PyVista for spatial visualization.

- **MLOps Infrastructure Deployment** (Jan – Feb 2025): Architected and integrated GitLab CI/CD pipelines into deep learning workflows for automated classification and nnU-Net segmentation, vastly improving model versioning and clinical reproducibility.
- **Anatomical 3D Segmentation** (Dec 2023 – Nov 2024): Implemented 3D segmentation architectures using nnU-Net on proprietary datasets to isolate the liver, kidneys, vascular structures, and pathology boundaries for downstream screening.
- **Multi-Label Landmark Detection** (May – Nov 2023): Built multi-label segmentation models using nnU-Net to detect critical kidney landmarks in CT and ultrasound, outperforming baseline models and enabling reliable cross-modal registration.

Machine Learning Engineer & Data Scientist

May 2020 – Dec 2022

- **Statistical Shape Modeling** (Nov 2022 – Jan 2023): Constructed Statistical Shape Models (SSMs) of kidney anatomy using the public KiTS19 dataset; applied Procrustes alignment, coherent point drift, and Principal Component Analysis (PCA) to quantify morphological variations.
- **Active Learning for Ultrasound Data** (Mar – Sep 2022): Formulated active learning strategies (via Lightly, modAL, and BaaL) for fetal ultrasound classification, reducing manual annotation requirements by 60% while maintaining baseline accuracy metrics.
- **Cross-Modal 3D Alignment** (Jul 2021 – Jan 2022): Developed a 3D CT-to-Ultrasound image registration framework utilizing the MINC Toolkit and SimpleElastix to guide computer-assisted minimally invasive surgeries.
- **Liver Mass Detection** (Oct 2020 – May 2021): Built an end-to-end 2D ultrasound object detection pipeline using DETR, YOLOv5, and VarifocalNet to classify solid vs. cystic liver masses. Enhanced final mean Average Precision (mAP) via test-time augmentation (TTA) and model ensembling.
- **Breast Tumor Classification** (May – Aug 2020): Trained ResNet and EfficientNet ensembles using the Catalyst framework to classify malignant vs. benign breast tumors, tracking experiments via MLflow and Hydra.

Awards & Scholarships

- **Chinese Government Scholarship (CSC)** : Highly competitive, merit-based full scholarship awarded for MSc studies. Selected through a rigorous national competition administered by Rwanda's Higher Education Council (HEC), 2017.

Technical Skills

- **Languages & tools:** python, PyTorch, Keras, Linux, Git/GitLab CI-CD, LaTeX, PostgreSQL, MLflow, Hydra, Catalyst, DVC, CVAT, 3D Slicer, SimpleElastix, Open3D.
- **Computer vision & deep learning:** 2D/3D Image Classification, Object Detection, 2D/3D Segmentation (nnU-Net), 2D/3D Image Registration, Statistical Shape Modeling, Active Learning, MLOps..